

Divide and conquer algorithms and Master theorem

Master Theorem

Consider the recursion $T(n) = aT(n/b) + f(n)$, where $a \geq 1, b > 1, f(n)$ positive (positive for all $n \in \mathbb{N}$). Then the following holds.

- If $f(n) = O(n^{\log_b a - \varepsilon})$ for some $\varepsilon > 0$, then $T(n) = \Theta(n^{\log_b a})$.
- if $f(n) = \Theta(n^{\log_b a})$, then $T(n) = \Theta(n^{\log_b a} \log n)$.
- if $f(n) = \Omega(n^{\log_b a + \varepsilon})$ for some $\varepsilon > 0$ AND
if $\exists c < 1 \exists N \in \mathbb{N}$, such that $af(n/b) \leq cf(n)$ holds for all $n \geq N$, then $T(n) = \Theta(f(n))$.

1. Show that Master theorem can not be applied for the following recursions.

- (a) $T(n) = 2^n T(n/2) + n^n$
- (b) $T(n) = 2T(n/2) + n/\log n$
- (c) $T(n) = 64T(n/8) - n^2 \log n$
- (d) $T(n) = T(n/2) + n(2 - \cos n)$

2. Solve these recursions by Master theorem (if possible).

- (a) $T(n) = 9T(n/3) + n$
- (b) $T(n) = T(2n/3) + 1$
- (c) $T(n) = 3T(n/4) + n \log n$
- (d) $T(n) = 2T(n/2) + n \log n$
- (e) $T(n) = 8T(n/2) + 1000n^2$
- (f) $T(n) = 2T(n/2) + 10n$
- (g) $T(n) = 2T(n/2) + n^2$
- (h) $T(n) = T(9n/10) + n$
- (i) $T(n) = 16T(n/4) + n^2$
- (j) $T(n) = 7T(n/2) + n^2$
- (k) $T(n) = 2T(n/4) + \sqrt{n}$

3. Give the recursion. Then solve the recursion by Master theorem.

- (a) binary search
- (b) merge sort
- (c) Determine silhouette of New York. (the i th building is described by $(a_i, b_i, h_i) \in \mathbb{R}^+ \times \mathbb{R}^+ \times \mathbb{R}^+$, base: from a_i to b_i ($a_i < b_i$), height: h_i for $1 \leq i \leq n$)
- (d) Maximal growth of a sequence $A[1..n]$, i.e. determine the pair of indices $1 \leq i < j \leq n$ with the maximum value of $A[j] - A[i]$.